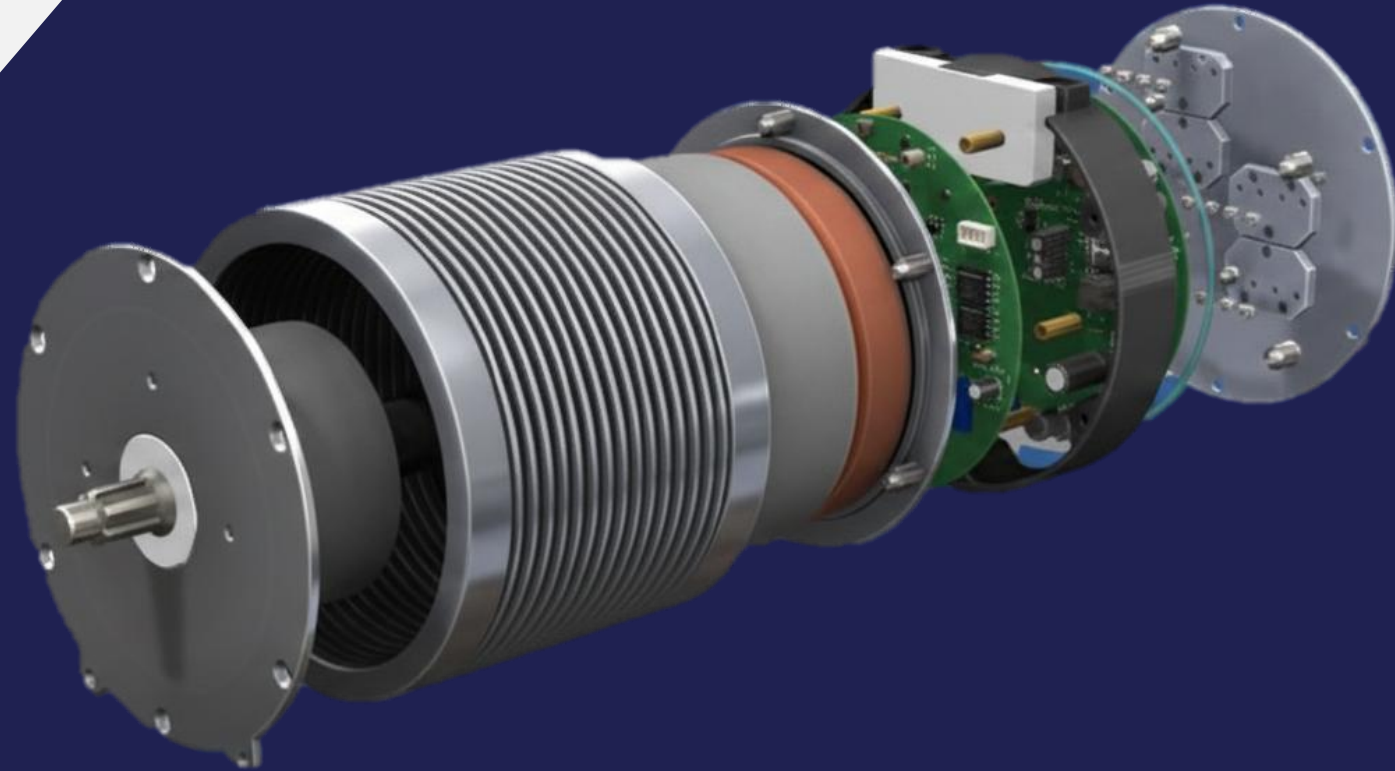




VOLTWORKS



Shaping the future of Motor &  
Controllers in Electric Vehicles

# About VoltWorks

VoltWorks is a technology first electric mobility company focused on developing and delivering high-performance drivetrain solutions for the Indian electric vehicle market.

We specialize in **Permanent Magnet Synchronous Motors (PMSM)** and **Field-Oriented Control (FOC)-based motor controllers**, designed specifically for two-wheelers, e-rickshaws, and light commercial vehicles. Our products are engineered for Indian road, weather, and load conditions—offering high torque, efficiency, and reliability at competitive costs.

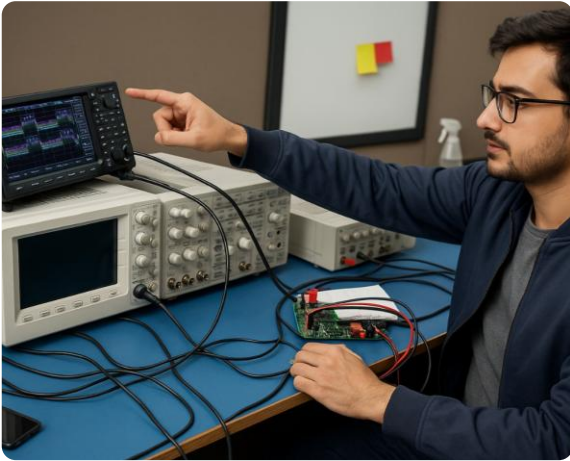
What sets VoltWorks apart is our ability to **offer deep**

**customization** for OEMs, enabling them to optimize vehicle performance for specific applications. Every component is designed, tested, and assembled in India, aligning with the **Make in India** and **Atmanirbhar Bharat** initiatives.

VoltWorks is backed by **STMicroelectronics**, one of the world's leading semiconductor companies, and proudly supported by **Citi Bank**, **IIT Kanpur**, and **IIT Indore**. At VoltWorks, we are not just building components—we are **powering the future of clean, smart, and reliable mobility**. Join us on our journey to electrify India, one vehicle at a time.



# Our Edge in EV Innovation



Our products undergo extensive lab and on-road testing across diverse terrains, loads, and environmental conditions. Every motor and controller is validated for long-term durability, thermal stability, and real-world performance.

All critical components — from PMSM motors to FOC controllers — are designed and developed by our in-house team. This ensures complete control over quality, performance tuning, and faster innovation cycles tailored to your application needs.



Designed, tested, and assembled in India, our solutions are optimized for Indian roads, climate, and customer expectations. We are proud contributors to the *Make in India* and *Atmanirbhar Bharat* movements, enabling self-reliance in EV technology.

With IP67-rated protection, optimize thermal performance, and high efficiency across operating ranges, VoltWorks' drivetrain systems deliver unmatched uptime and reliability - even under the toughest operating conditions.





# Product & Services



## Motor & Controller

Electric PMSM Motors  
FOC Based Motor Controllers  
Integrated Drivetrain Systems



## EV Electronics

Cluster & Displays  
DCDC Converters  
Power Distribution Junction Box



System Design  
FEA and Simulation  
Technical Support

## Engineering Consulting

Prototyping  
Testing & Validation  
Manufacturing Support

## Product Development



# Motor & Controllers

## L3 Application

PMSM L3 Drivetrain  
1000W – 2000W



## L5 Application

PMSM L3 Drivetrain  
2000W – 4000W



## L2 Application

PMSM L3 Drivetrain  
800W – 2000W



## Features of VoltWorks

**24/7 SERVICE**

Waterproof Motor

Overload Protection

Hairpin Winding

External Sensor

# PMSM Motor – 120S – 60H – 1200W

Waterproof Motor

20% High Torque & Range

High Speed ~ 35 kmph

Overload Protection

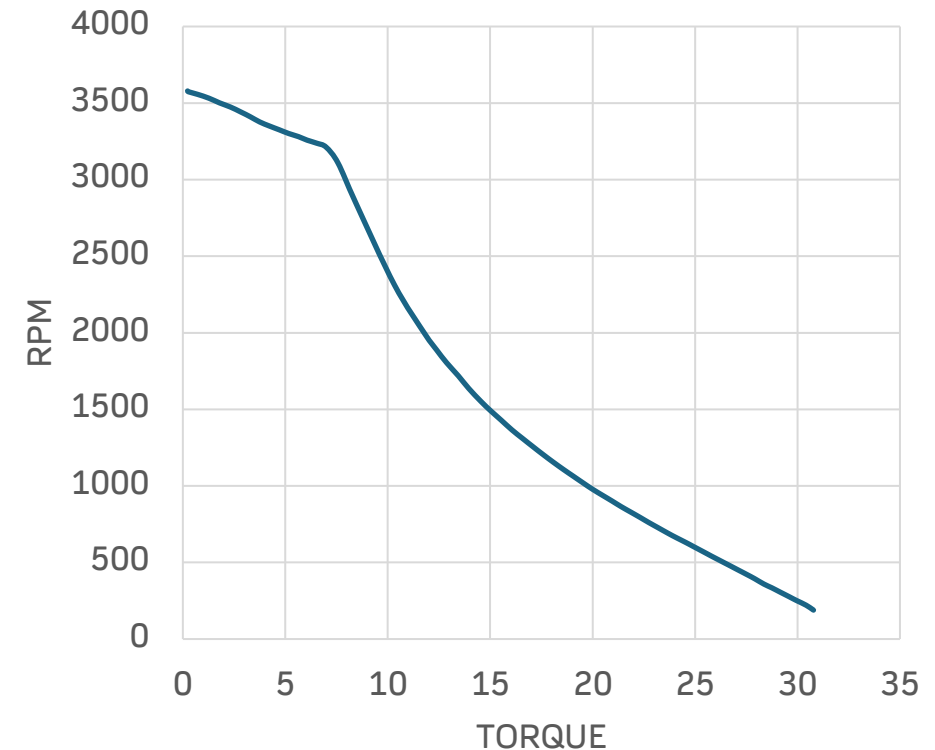
|                  |                        |
|------------------|------------------------|
| Rated Voltage    | 48 V   60 V            |
| Continuous Power | 1250 W                 |
| Nominal Torque   | 8 Nm                   |
| Peak Torque      | 30 Nm                  |
| Peak RPM         | 3550                   |
| Peak Efficiency  | > 93%                  |
| IP Protections   | IP66                   |
| Cooling          | Forced Fan Air Cooling |
| Gradability      | 18%                    |
| GVW              | 700 kg                 |
| Application      | L3 E-Passenger         |
| Dimensions       | OD: 142 mm; L: 155 mm  |



APPLICATION IN  
L3 PASSENGER



TORQUE VS RPM



# PMSM Motor – 120S – 70H – 1500W

Waterproof Motor

20% High Torque & Range

High GVW ~ 1000 kg

Overload Protection

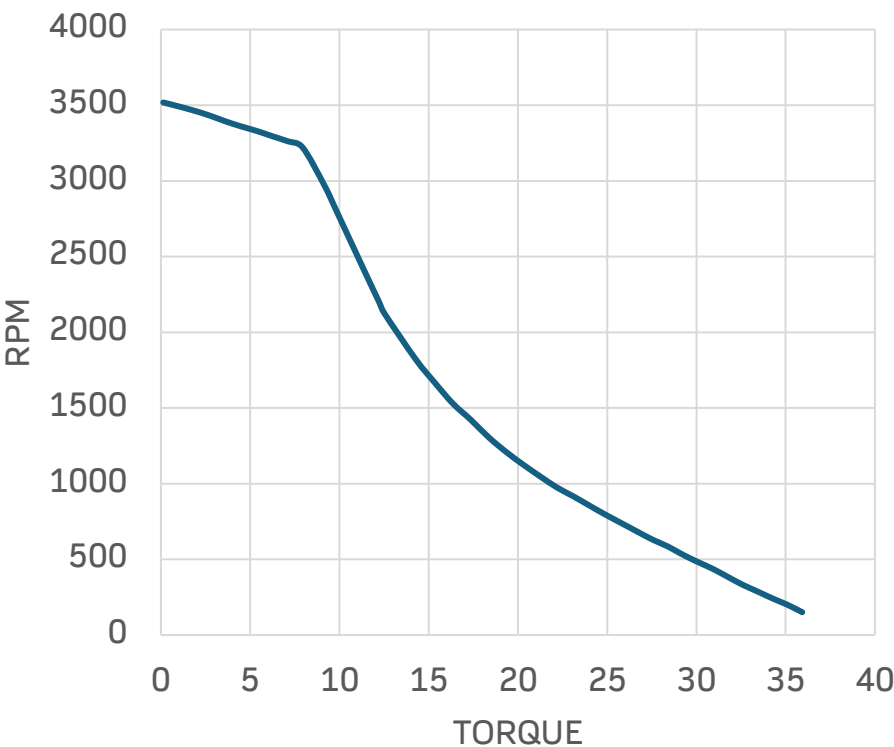
|                  |                        |
|------------------|------------------------|
| Rated Voltage    | 48 V   60 V            |
| Continuous Power | 1500 W                 |
| Nominal Torque   | 10 Nm                  |
| Peak Torque      | 37 Nm                  |
| Peak RPM         | 3450                   |
| Peak Efficiency  | > 93%                  |
| IP Protections   | IP65 (90%)   IP66      |
| Cooling          | Forced Fan Air Cooling |
| Gradability      | 18%                    |
| GVW              | 1000 kg                |
| Application      | L3 E-Loader            |
| Dimensions       | OD: 142 mm; L: 165 mm  |



APPLICATION IN  
L3 LOADER



TORQUE VS RPM



# PMSM Motor – 140S – 65H – 2000W

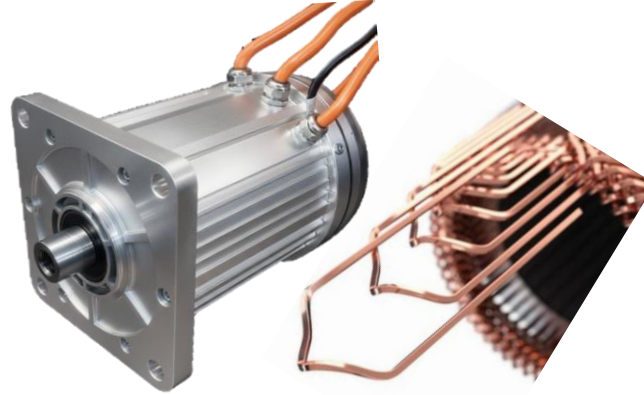
Hairpin Winding Tech

High Torque > 75 Nm

High Gradeability 20%

External Position Sensor

|                  |                        |
|------------------|------------------------|
| Rated Voltage    | 48V   60V   72 V       |
| Continuous Power | 2000 W                 |
| Nominal Torque   | 12 Nm                  |
| Peak Torque      | 68 Nm                  |
| Peak RPM         | 4800   5400   6000     |
| Peak Efficiency  | > 92%                  |
| IP Protections   | IP66                   |
| Cooling          | Forced Fan Air Cooling |
| Gradability      | 20%                    |
| GVW              | 1000 kg                |
| Application      | Passenger / Cargo      |
| Dimensions       | OD: 165 mm; L: 195 mm  |

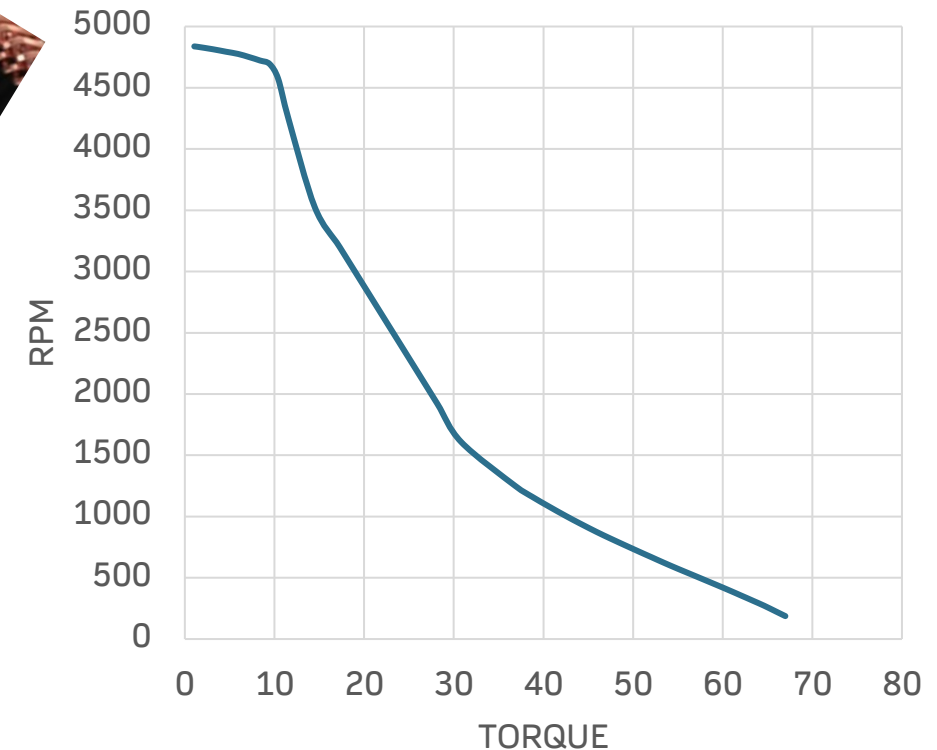


HAIRPIN TECHNOLOGY INSIDE

APPLICATION IN  
PASSENGER / CARGO



TORQUE VS RPM





# PMSM Motor – 140S – 75H – 3000W

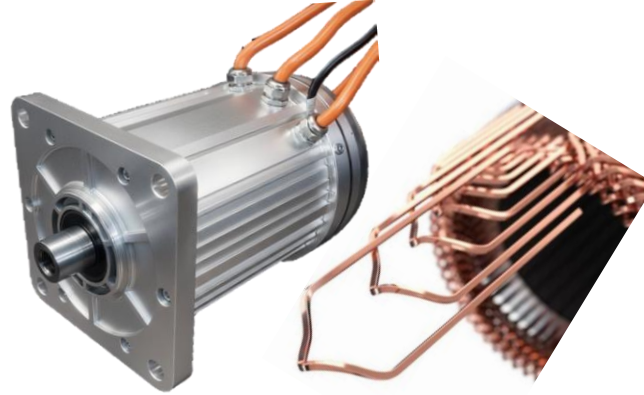
Hairpin Winding Tech

High Torque > 75 Nm

High Gradeability 20%

External Sensor

|                  |                        |
|------------------|------------------------|
| Rated Voltage    | 48V   60V   72 V       |
| Continuous Power | 3000 W                 |
| Nominal Torque   | 16 Nm                  |
| Peak Torque      | 75 Nm                  |
| Peak RPM         | 5000   5600   6200     |
| Peak Efficiency  | > 92%                  |
| IP Protections   | IP66                   |
| Cooling          | Forced Fan Air Cooling |
| Gradability      | 20%                    |
| GVW              | 1200 kg                |
| Application      | L5 Passenger           |
| Dimensions       | OD: 165 mm; L: 205 mm  |

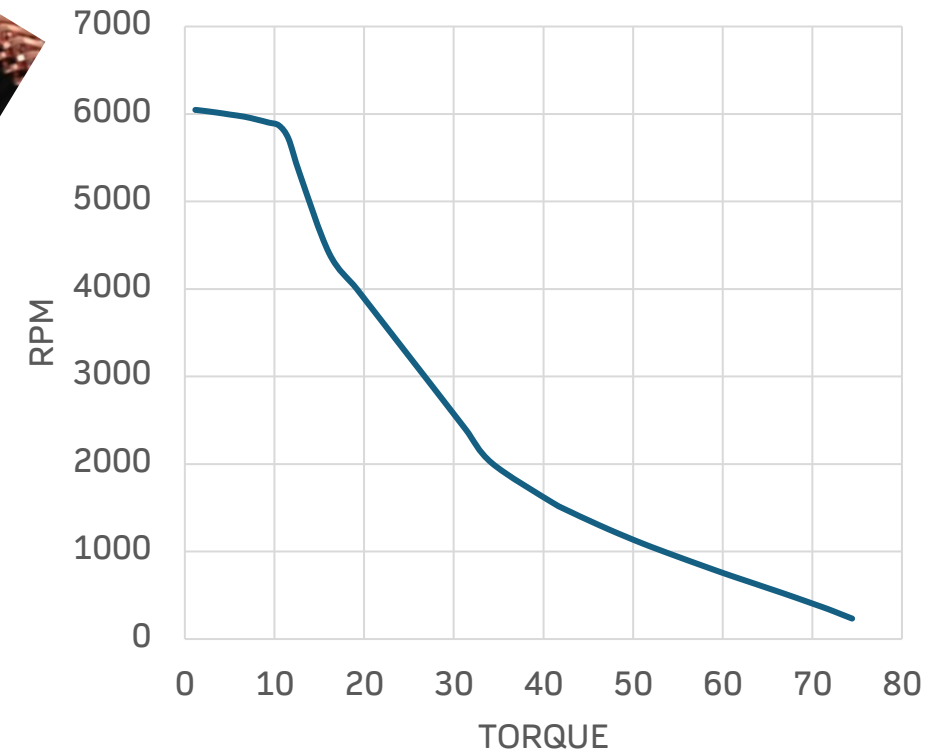


HAIRPIN TECHNOLOGY INSIDE

APPLICATION IN  
L5 PASSENGER / CARGO



TORQUE VS RPM



# PMSM Motor – 140S – 85H – 4000W

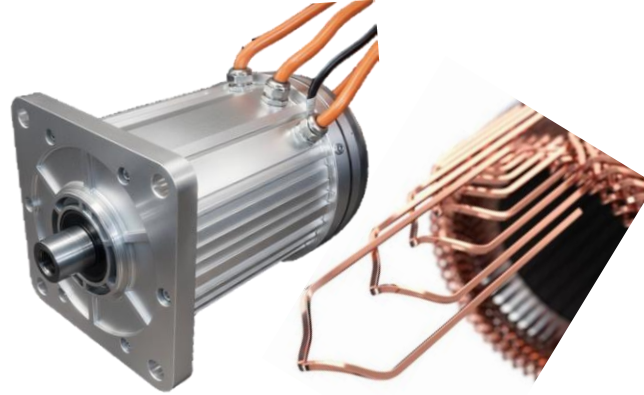
Hairpin Winding Tech

High Torque > 80 Nm

High Gradeability 20%

External Sensor

|                  |                        |
|------------------|------------------------|
| Rated Voltage    | 48V   60V   72 V       |
| Continuous Power | 4000 W                 |
| Nominal Torque   | 20 Nm                  |
| Peak Torque      | 80 Nm                  |
| Peak RPM         | 5400   6000   6700     |
| Peak Efficiency  | > 92%                  |
| IP Protections   | IP66                   |
| Cooling          | Forced Fan Air Cooling |
| Gradability      | 20%                    |
| GVW              | 1500 kg                |
| Application      | L5 Loader              |
| Dimensions       | OD: 165 mm; L: 215 mm  |

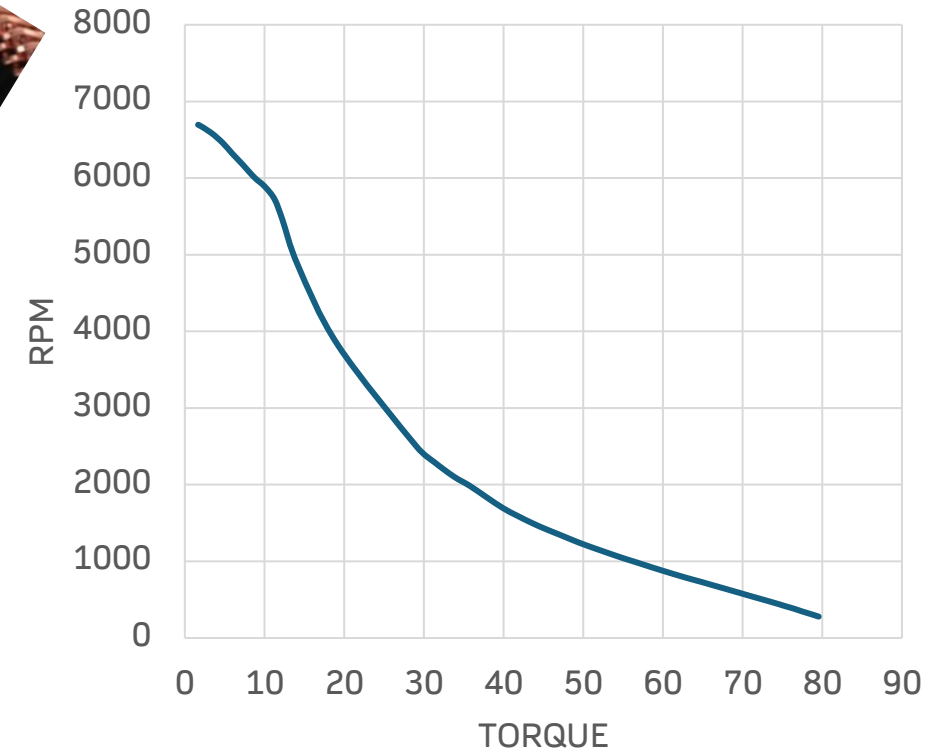


HAIRPIN TECHNOLOGY INSIDE

APPLICATION IN  
L5 CARGO



TORQUE VS RPM



# PMSM Motor – 10”S – 27H – 1000W

Thicker Magnet ~ 2.3 mm

Oil Cooled Motor

Dual Hall Sensor

Waterproof Motor

| Parameters     | T(Nm) | N(RPM) | Pout(W) | U(V) | I(A) | Pin(W) | Efficiency (%) |
|----------------|-------|--------|---------|------|------|--------|----------------|
| No Load        | 0.0   | 537    | 0.0     | 48   | 2.5  | 120    | 0.0            |
| Max Efficiency | 23.7  | 472    | 1169    | 48   | 27.8 | 1336   | 87.5           |
| Max Pout       | 26.4  | 437    | 1201    | 48   | 28.9 | 1388   | 86.6           |
| Max Torque     | 73.3  | 28     | 212     | 48   | 28.8 | 1383   | 15.2           |
| End Point      | 56.7  | 0.2    | 1.0     | 48   | 17.6 | 849    | 0.1            |



APPLICATION IN  
L2 SCOOTERS



| Wheel Diameter (In) | Voltage     | Stack Length (mm) | Magnet Thick (mm) | Peak Current | Brake Type  |
|---------------------|-------------|-------------------|-------------------|--------------|-------------|
| 10 inch             | 48 V / 60 V | 27mm              | 2.3 mm            | 32 A         | Disc / Drum |

# PMSM Motor – 10”S – 30H – 1200W

Thicker Magnet ~ 2.3 mm

Oil Cooled Motor

Dual Hall Sensor

Waterproof Motor

| Parameters     | T(Nm) | N(RPM) | Pout(W) | U(V) | I(A) | Pin(W) | Efficiency (%) |
|----------------|-------|--------|---------|------|------|--------|----------------|
| No Load        | 0.0   | 670    | 0.00    | 60   | 2.7  | 165    | 0.0            |
| Max Efficiency | 24.7  | 589    | 1519    | 60   | 28.5 | 1714   | 88.6           |
| Max Pout       | 27.8  | 532    | 1538    | 60   | 29.1 | 1747   | 88.1           |
| Max Torque     | 85.0  | 33     | 292     | 60   | 28.6 | 1720   | 16.8           |
| End Point      | 59.0  | 0.0    | 0.0     | 60   | 3.5  | 211    | 0.0            |



APPLICATION IN  
L2 SCOOTERS



| Wheel Diameter (In) | Voltage     | Stack Length (mm) | Magnet Thick (mm) | Peak Current | Brake Type  |
|---------------------|-------------|-------------------|-------------------|--------------|-------------|
| 10 inch             | 48 V / 60 V | 30mm              | 2.3 mm            | 35 A         | Disc / Drum |



# PMSM Motor – 12”S – 27H – 1200W

Thicker Magnet ~ 2.3 mm

Oil Cooled Motor

Dual Hall Sensor

Waterproof Motor

| Parameters     | T(Nm) | N(RPM) | Pout(W) | U(V) | I(A) | Pin(W) | Efficiency (%) |
|----------------|-------|--------|---------|------|------|--------|----------------|
| No Load        | 0.0   | 554.0  | 0.0     | 60   | 2.5  | 150    | 0.0            |
| Max Efficiency | 29.0  | 487.0  | 1476    | 60   | 28.0 | 1682   | 87.7           |
| Max Pout       | 33.0  | 446.0  | 1531    | 60   | 29.3 | 1762   | 86.9           |
| Max Torque     | 98.0  | 22.0   | 222     | 60   | 28.7 | 1728   | 12.7           |
| End Point      | 86.0  | 0.1    | 0.8     | 60   | 17.6 | 1058   | 0.1            |



APPLICATION IN  
L2 SCOOTERS



| Wheel Diameter (In) | Voltage     | Stack Length (mm) | Magnet Thick (mm) | Peak Current | Brake Type  |
|---------------------|-------------|-------------------|-------------------|--------------|-------------|
| 12 inch             | 48 V / 60 V | 27mm              | 2.3 mm            | 35 A         | Disc / Drum |

# PMSM Controller – L3 Series

|                   |                             |
|-------------------|-----------------------------|
| Rated Voltage     | 48 V   60V                  |
| Peak DC Current   | 60 A   70 A                 |
| Continuous Arms   | 40 A   50 A                 |
| Peak Arms         | 150 A   180 A               |
| Power Rating      | 1200 W   1500 W             |
| Peak Efficiency   | > 95%                       |
| IP Protections    | IP54                        |
| Cooling           | Air Cooled                  |
| Compatible Sensor | Hall Sensor                 |
| Protection        | Overload (Temp Sensor)      |
| Application       | L3 E-Rickshaw               |
| Features          | Anti Slip; SW Customization |



## APPLICATION IN L3 PASSENGER / CARGO



## Features

- ❖ Field-Orient Control (FOC) with optimized motor parameters
- ❖ Precise Motor torque control
- ❖ Compact, Light weight with wide Operational Voltage Range
- ❖ Compatible with Hall Sensor
- ❖ Flexibility in Customizing software

## Drive train performance optimizations

- ❖ VoltWorks controllers comes with optimized controls
- ❖ Maximises torque output
- ❖ Smooth torque changeover and ramps

# PMSM Controller – L5 Series

|                   |                             |
|-------------------|-----------------------------|
| Rated Voltage     | 48 V   60V   72V            |
| Peak DC Current   | 80 A   90 A   100 A         |
| Continuous Arms   | 80 A                        |
| Peak Arms         | 200 A                       |
| Power Rating      | 2000 W   3000 W   4000 W    |
| Peak Efficiency   | > 95%                       |
| IP Protections    | IP67                        |
| Cooling           | Air Cooled                  |
| Compatible Sensor | Rotary Transformer          |
| Protection        | Overload (Temp Sensor)      |
| Application       | L5 3-Wheeler                |
| Features          | Anti Slip; SW Customization |



## APPLICATION IN L5 PASSENGER / CARGO



## Features

- ❖ Field-Orient Control (FOC) with optimized motor parameters
- ❖ Precise Motor torque control
- ❖ Compact, Light weight with wide Operational Voltage Range
- ❖ Compatible with Hall Sensor
- ❖ Flexibility in Customizing software

## Drive train performance optimizations

- ❖ VoltWorks controllers comes with optimized controls
- ❖ Maximises torque output
- ❖ Smooth torque changeover and ramps

# PMSM Controller – L2 Series

|                   |                             |
|-------------------|-----------------------------|
| Rated Voltage     | 48 V   60V   72V            |
| Peak DC Current   | 30A   35A   40A             |
| Continuous Arms   | 35A   40 A   50 A           |
| Peak Arms         | 50 A                        |
| Power Rating      | 800 W   1200 W   1500W      |
| Peak Efficiency   | > 95%                       |
| IP Protections    | IP54                        |
| Cooling           | Air Cooled                  |
| Compatible Sensor | Hall Sensor                 |
| Protection        | Overload (Temp Sensor)      |
| Application       | L2 Scooters                 |
| Features          | Anti Slip; SW Customization |



## APPLICATION IN L2 SCOOTERS



## Features

- ❖ Field-Orient Control (FOC) with optimized motor parameters
- ❖ Over & under voltage protection
- ❖ Compact, Lightweight with wide Operational Voltage Range
- ❖ Three operation modes and over current protection
- ❖ Hill holds, vector control, cruise control

## Drive train performance optimizations

- ❖ VoltWorks controllers comes with optimized controls
- ❖ Maximises torque output
- ❖ Smooth torque changeover and ramps



# Our Customers & Partners

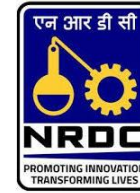


STARTUP  
INCUBATION AND  
INNOVATION  
CENTRE  
IIT KANPUR

**citibank**



**IIT  
Delhi**



**NRDC**



**Balan**



# Board of Directors



**IIT** INDIAN  
INSTITUTE OF  
TECHNOLOGY  
BANARAS HINDU UNIVERSITY



Himanshu Sahu  
Co-Founder & CEO

Experience in leading the teams and launching the product in market



**IIT**  
**Delhi**

**oerlikon**

Devendra Chaudhary  
Co-Founder & CBO

17+ years of experience in building companies in HW, SW products primarily in automotive



**IIT** INDIAN  
INSTITUTE OF  
TECHNOLOGY  
BANARAS HINDU UNIVERSITY



Rishabh Singh  
Co-Founder & CTO

A techy in building products from scratch to commercialization.

At VoltWorks, we are driven by a vision to redefine the future of mobility in India through innovation, reliability, and sustainability. Together with our partners, we are building the foundation for a cleaner, more connected, and electrified tomorrow—proudly made in India, for the world.

# Calendar 2026

## January

| S  | M  | T  | W  | T  | F  | S  |
|----|----|----|----|----|----|----|
|    |    |    |    | 1  | 2  | 3  |
| 4  | 5  | 6  | 7  | 8  | 9  | 10 |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 |
| 18 | 19 | 20 | 21 | 22 | 23 | 24 |
| 25 | 26 | 27 | 28 | 29 | 30 | 31 |

## February

| S  | M  | T  | W  | T  | F  | S  |
|----|----|----|----|----|----|----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  |
| 8  | 9  | 10 | 11 | 12 | 13 | 14 |
| 15 | 16 | 17 | 18 | 19 | 20 | 21 |
| 22 | 23 | 24 | 25 | 26 | 27 | 28 |

## March

| S  | M  | T  | W  | T  | F  | S  |
|----|----|----|----|----|----|----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  |
| 8  | 9  | 10 | 11 | 12 | 13 | 14 |
| 15 | 16 | 17 | 18 | 19 | 20 | 21 |
| 22 | 23 | 24 | 25 | 26 | 27 | 28 |
| 29 | 30 | 31 |    |    |    |    |

## April

| S  | M  | T  | W  | T  | F  | S  |
|----|----|----|----|----|----|----|
|    |    |    | 1  | 2  | 3  | 4  |
| 5  | 6  | 7  | 8  | 9  | 10 | 11 |
| 12 | 13 | 14 | 15 | 16 | 17 | 18 |
| 19 | 20 | 21 | 22 | 23 | 24 | 25 |
| 26 | 27 | 28 | 29 | 30 |    |    |

## May

| S  | M  | T  | W  | T  | F  | S  |
|----|----|----|----|----|----|----|
|    |    |    |    |    | 1  | 2  |
| 3  | 4  | 5  | 6  | 7  | 8  | 9  |
| 10 | 11 | 12 | 13 | 14 | 15 | 16 |
| 17 | 18 | 19 | 20 | 21 | 22 | 23 |
| 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 31 |    |    |    |    |    |    |

## June

| S  | M  | T  | W  | T  | F  | S  |
|----|----|----|----|----|----|----|
|    | 1  | 2  | 3  | 4  | 5  | 6  |
| 7  | 8  | 9  | 10 | 11 | 12 | 13 |
| 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 |
| 28 | 29 | 30 |    |    |    |    |

## July

| S  | M  | T  | W  | T  | F  | S  |
|----|----|----|----|----|----|----|
|    |    |    | 1  | 2  | 3  | 4  |
| 5  | 6  | 7  | 8  | 9  | 10 | 11 |
| 12 | 13 | 14 | 15 | 16 | 17 | 18 |
| 19 | 20 | 21 | 22 | 23 | 24 | 25 |
| 26 | 27 | 28 | 29 | 30 | 31 |    |

## August

| S  | M  | T  | W  | T  | F  | S  |
|----|----|----|----|----|----|----|
|    |    |    |    |    |    | 1  |
| 2  | 3  | 4  | 5  | 6  | 7  | 8  |
| 9  | 10 | 11 | 12 | 13 | 14 | 15 |
| 16 | 17 | 18 | 19 | 20 | 21 | 22 |
| 23 | 24 | 25 | 26 | 27 | 28 | 29 |
| 30 | 31 |    |    |    |    |    |

## September

| S  | M  | T  | W  | T  | F  | S  |
|----|----|----|----|----|----|----|
|    |    | 1  | 2  | 3  | 4  | 5  |
| 6  | 7  | 8  | 9  | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 | 17 | 18 | 19 |
| 20 | 21 | 22 | 23 | 24 | 25 | 26 |
| 27 | 28 | 29 | 30 |    |    |    |

## October

| S  | M  | T  | W  | T  | F  | S  |
|----|----|----|----|----|----|----|
|    |    |    |    | 1  | 2  | 3  |
| 4  | 5  | 6  | 7  | 8  | 9  | 10 |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 |
| 18 | 19 | 20 | 21 | 22 | 23 | 24 |
| 25 | 26 | 27 | 28 | 29 | 30 | 31 |

## November

| S  | M  | T  | W  | T  | F  | S  |
|----|----|----|----|----|----|----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  |
| 8  | 9  | 10 | 11 | 12 | 13 | 14 |
| 15 | 16 | 17 | 18 | 19 | 20 | 21 |
| 22 | 23 | 24 | 25 | 26 | 27 | 28 |
| 29 | 30 |    |    |    |    |    |

## December

| S  | M  | T  | W  | T  | F  | S  |
|----|----|----|----|----|----|----|
|    |    | 1  | 2  | 3  | 4  | 5  |
| 6  | 7  | 8  | 9  | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 | 17 | 18 | 19 |
| 20 | 21 | 22 | 23 | 24 | 25 | 26 |
| 27 | 28 | 29 | 30 | 31 |    |    |



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CONNECTED MOBILITY



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